PH.48111 **FALL 2026**

Autonomous Theory Construction Laboratory

Lecture 2: Designing Reference Data

Garam Yun 2026-09-07–2026-09-13

Overview

Reference data are specified by values, units, likelihoods, instrument metadata, and version status.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$O_a = (y_a, unit, L_a, I_a, v_a, status_a)$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$R_a = (y_a - T_a)/\sigma_a$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$L_R = -2 \sum_a \ln p_a(y_a | T_a)$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Yun, ch. 2
- Registry Standard ORS-15
- RLP-12 §4